SortIQ — Next-Generation Smart Waste Classification System

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Abstract

SortIQ is an AI-based waste classification system that detects recyclable materials in real time using computer vision. The system identifies Glass, Plastic, Metal, and Paper through a live camera feed and provides visual feedback and recycling guidance. It combines YOLOv8 object detection with MobileNetV2 material classification to analyze objects and record confirmed scans into history and analytics dashboards. The frontend interface is deployed on Vercel, while the AI backend currently runs locally due to cloud memory limitations.

Method

Figure 1: System Design

The goal of SortIQ is to develop a smart system that assists users in correctly identifying recyclable waste materials.

The detection pipeline includes:

- Object Detection – Objects are detected in real time using YOLOv8.
- Material Classification – Detected objects are analyzed using MobileNetV2 to classify their material type.
- Verification Stage – Additional color and texture checks improve classification reliability.
- Stability Check – Detections must remain consistent across multiple frames before confirmation.
- Data Recording – Confirmed detections are stored automatically in the History and Analytics dashboards.

Figure 2: Result Recommendation

Result

SortIQ successfully detects recyclable materials in real time and visualizes them with category colors and confidence levels. Confirmed scans are automatically recorded, allowing users to monitor recycling activity through the analytics dashboard

Discussion

The system demonstrates how computer vision and artificial intelligence can support sustainable waste management. Combining object detection with material classification improves accuracy while preventing incorrect detections.

Conclusion

SortIQ provides an intelligent and interactive solution for waste classification. The system shows strong potential for future applications in smart recycling systems and AI-assisted environmental technologies